



Detecting Dirty Dish Culprits

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Introduction

- Outputs the time stamps when someone leaves behind dirty dishes



Related Work

- DISH-O-TRON uses a computer vision AI model to detect dirty dishes in communal kitchen sinks
- Object detection focusing on the dishes themselves



[1] "Dish-O-tron – no more dirty dishes thanks to ai," codecentric AG,
<https://www.codecentric.de/wissens-hub/blog/dish-o-tron-no-more-dirty-dishes-ai> (accessed Dec. 9, 2024).

Method

- Process data by drawing Classification bounding boxes



Method

- Initialize the model by loading the pre-trained weights
 - Adam Optimizer, LR = 0.001667, momentum = 0.9, 20 epochs, Cross Entropy Loss
- Model is split into two parts: frame by frame object detection and continuity detection

```
from ultralytics import YOLO
from datetime import datetime

import cv2
import time
# Import other required libraries

# Initialize the YOLO model by loading the pre-trained weights from 'yolov8s.pt'
model = YOLO('best (3).pt')

source = "pnm_test_footage_again.mp4"

cap = cv2.VideoCapture(source)

label_count = {}
consecutive_labels = []

fps = int(cap.get(cv2.CAP_PROP_FPS))
def get_timestamp(frame_idx):
    seconds = frame_idx / fps
    return str(datetime.utcfromtimestamp(seconds).strftime('%H:%M:%S'))

frame_idx = 0
non_con = 0
path_to_start = None
non_consecutive_count = 0
wash_counter = 0
prev_cls = "Walking"

ret, frame = cap.read()
```

Method

- Make predictions
- Determine if fake washing occurred (someone dropped off a dirty dish)

```
#predict
results = model(source, stream=True, show_conf=True, show_labels=True, show_boxes=True, save=True, conf=.5, save_txt=None)
file = open("predicted_labels.txt", 'w')
labels = []
for r in results:
    boxes = r.boxes

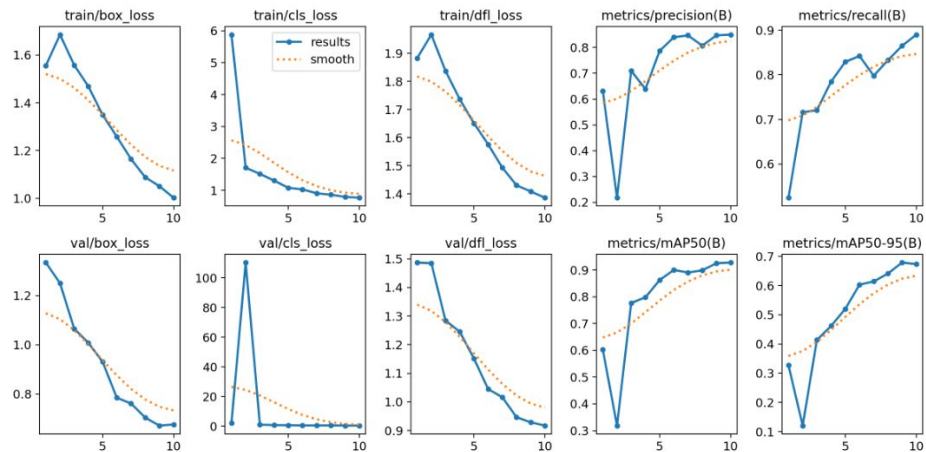
    for idx, prediction in enumerate(boxes.xywhn):
        cls = int(boxes.cls[idx].item())
        class_name = model.names[cls]
        labels.append(class_name)
        file.write(f"{class_name} {cls} {prediction[0].item()} {prediction[1].item()} {prediction[2].item()} {prediction[3].item()}\n")

    masks = r.masks
    prob = r.probs

prev_path_to_start = 0
path_to_start = 0
for label in labels:
    frame_idx += 1
    if label == "Washing":
        if non_con >= 10:
            path_to_start = frame_idx
            non_con = 0
            wash_counter += 1
        else:
            non_con += 1
            if wash_counter >= 1 and wash_counter <= 20 and path_to_start != prev_path_to_start:
                print(f"Fake washing detected at frame {path_to_start}")
                prev_path_to_start = path_to_start
                wash_counter = 0
    prev_cls = label
```

Experiments

- Train, validation, and test split in a ratio of 70/20/10
- Tested our model on all of the data to check if it worked robustly for our action classification until satisfactory results were achieved, adding more trained data between results
- Final mAP50 score of .96



Conclusion

- Our model was able to detect if people were washing their dishes with high fidelity
- If found in washing position for an insufficient amount of time, it will flag the person as dropping of a dirty dish



Fake washing detected, returning frame 79 for review